

CHEN CHEN

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EDUCATION

- Tsinghua University (THU), Beijing, China** Sept. 2024 – Jun. 2027 (expected)
- *Master of Science in Control Science and Engineering*
 - Research Interests: Human–Robot Interaction, Medical Robotics
- Tsinghua University (THU), Beijing, China** Sept. 2020 – Jun. 2024
- *Bachelor of Science in Automation*
 - GPA: 3.96/4.0 (Sophomore and Junior year), 3.75/4.0 (overall)

PUBLICATION

[6] **Robotic In-Hand Manipulation for Large-Range Precise Object Movement: The RGMC Champion Solution** 

Mingrui Yu, Yongpeng Jiang, **Chen Chen**, Yongyi Jia and Xiang Li,
in *IEEE Robotics and Automation Letters*, May 2025

[5] **Visual Attention Based Cognitive Human–Robot Collaboration for Pedicle Screw Placement in Robot-Assisted Orthopedic Surgery** 

Chen Chen, Qikai Zou, Yuhang Song, Mingrui Yu, Senqiang Zhu, Shiji Song and Xiang Li,
in *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*

[4] **Multi-Modal Interaction Control of Ultrasound Scanning Robots with Safe Human Guidance and Contact Recovery** 

Xiangjie Yan, Shaqi Luo, Yongpeng Jiang, Mingrui Yu, **Chen Chen**, Senqiang Zhu, Gao Huang, Shiji Song and Xiang Li, in *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*

[3] **Independence in the Home: A Wearable Interface for a Person with Quadriplegia to Teleoperate a Mobile Manipulator**     **Best Paper Award in Systems**

Akhil Padmanabha, Janavi Gupta, **Chen Chen**, Jehan Yang, Vy Nguyen, Douglas J Weber, Carmel Majidi and Zackory Erickson, in *2024 ACM/IEEE International Conference on Human-Robot Interaction (HRI)*

[2] **A Complementary Framework for Human–Robot Collaboration With a Mixed AR–Haptic Interface**  

Xiangjie Yan, Yongpeng Jiang, **Chen Chen**, Leiliang Gong, Ming Ge, Tao Zhang and Xiang Li, *IEEE Transactions on Control Systems Technology*, Jan 2024

[1] **Adaptive Vision-Based Control of Redundant Robots with Null-Space Interaction for Human–Robot Collaboration**  

Xiangjie Yan, **Chen Chen** and Xiang Li, in *2022 International Conference on Robotics and Automation (ICRA)*

HONORS AND AWARDS (SELECTED)

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| • Outstanding Graduate , Department of Automation, Tsinghua University | Jun. 2024 |
| • Outstanding Graduate in Beijing , Beijing Municipal Education Commission | Jun. 2024 |
| • Jiang Nanxiang Scholarship , Tsinghua University
The second-highest scholarship at Tsinghua, awarded to only 40 undergrads annually. | Oct. 2023 |
| • China National Scholarship , Ministry of Education of the People's Republic of China
Top 0.1%, highest scholarship given by the Chinese government. | Dec. 2022 |
| • Outstanding Project of Student Research Training (SRT) Program , Tsinghua University
Top 5% of all SRT projects at Tsinghua University. | Dec. 2022 |

- Most Elegant Solution, 9th **Robotic Grasping and Manipulation Competition** at ICRA 2024 May. 2024
1st Place in RGMC In-Hand Manipulation Track.
- First Prize in the **Beijing Challenge Cup** May. 2023
- Second Prize in the 23rd **Electronic Design Competition**, Tsinghua University Dec. 2021
- Second Prize in the 15th **Intelligent Vehicle Competition**, Tsinghua University Apr. 2021
- First Prize in the 33rd **Chinese Chemistry Olympiad** Sept. 2019

RESEARCH EXPERIENCE

Intelligent Robotic Manipulation Lab, Dept. of Automation, Tsinghua University Apr. 2021 – Present

Research Assistant Advisor: Xiang Li

- Designed a novel Augmented Reality (AR) interface for interaction with robots' null space. [1]
 - Enabled a UR5 robot to carry out tasks with an uncalibrated camera while interacting with humans via the AR interface to deal with unforeseen changes.
 - Ensured efficient and safe collaboration without affecting the robot end-effector's main task.
- Proposed a complementary framework for human-robot collaboration with an AR-haptic interface. [2]
 - Enabled a Franka robot to carry out a picking task using a vision-based adaptive controller while the human expert supervises and manipulates the robot's null space to avoid collisions.
 - Extended the interface proposed in [1] by adding a haptic device, which allows the robot to learn the expert's demonstration with dynamic movement primitives (DMP) in a placing task.
- Proposed a novel multi-modal control scheme for ultrasound scanning robots. [4]
 - Achieved automatic switching between different control modes smoothly, depending on the doctor's actions and changes in the environment, such as the movement of the patient's body.
 - Combined the advantages of the doctor's experience/knowledge and the robot's autonomous ability, allowing the doctor to intervene safely at any time while maximizing the robot's scanning ability.
 - Developed a perception system based on Azure Kinect, which can recognize the doctor's actions and generate the movement trajectory of the ultrasound probe on the patient's neck.
- Developed a visual attention-based cognitive human-robot collaboration system for surgical robots. [5]
 - Proposed a novel HRI scheme for safety-critical tasks like surgery.
 - Developed a shared interaction controller for the robot.

Robotic Caregiving and Human Interaction Lab, RI, Carnegie Mellon University Apr. 2023 – Jun. 2024

Research Intern Advisor: Zackory Erickson

- Developed a shared control scheme for a assistive robot to help people with quadriplegia to teleoperate a mobile manipulator. [3]
- Conducted a week-long human study to evaluate the system's usability and performance.

Medical Cosmetic Robot Project, Tsinghua and Peking Union Medical College Sept. 2022 – Mar. 2024

Research Assistant Advisor: Gangtie Zheng, Xiao Long

- Developed a surgery robot for mesotherapy, a non-invasive cosmetic treatment.
- Implemented admittance control scheme on UR5e and developed a point cloud registration method to obtain geometric relationships between the patient and a previously obtained high-precision facial model.

SKILLS

Programming C/C++, Python, C#, MATLAB, Julia, Rust, Java

Tools ROS, ROS 2, L^AT_EX, Unity3D, Docker, KiCAD, SolidWorks, Blender

Platforms UR5(e), Franka Emika Panda, Stretch, Unitree Go1, HoloLens 2, Omega 3, and so on

Languages Chinese (native), English (fluent, GRE 156+170)

PROFESSIONAL ACTIVITIES

Reviewer of RSS, HRI, IROS, TASE.

Open source contribution: PointCloudLibrary/pcl, UniversalRobots/Universal_Robots_ROS2_Driver